Perturbations of the Einstein Cylinder in the Newman Penrose Formalism

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Abstract

The Newman Penrose formalism is a useful tool for analysing gravitational waves. We present a brief overview of this framework, and apply it to the Einstein Cylinder. This is the first attempt at studying gravitational waves on this manifold with the Newman Penrose formalism. Obtaining a solution on this conformally flat space-time would allow us to describe solutions on other conformally flat space-times. Thus, providing a deeper understanding of gravitational waves on manifolds with different asymptotic structures. Unfortunately, the perturbation decoupling procedure given by Teukolsky [1] does not work for this background metric. We discuss why this approach presents unforeseen complications, and suggest some more promising avenues for research.

1 Introduction

The discovery of the general theory of relativity by Einstein in 1915 [2] was a landmark achievement in our understanding of gravity. A key prediction of this theory is the existence of gravitational waves [3] which have since been observed by LIGO [4]. One can study the behaviour of gravitational waves in a space-time by considering perturbations around a background solution to the Einstein Field Equations. For somewhat complicated background metrics, the process can become quite involved. In 1957, Regge and Wheeler successfully applied this method to a Schwarzschild Black Hole [5]. However, until Teukolsky in 1973, perturbations around the metric of a Kerr Black Hole had not been calculated. The Newman Penrose formalism [6] is well suited for analysing such geometries as they are Petrov Type D [1]. This means there exists some natural choice of basis, or spin-frame, which simplifies calculations. Much of the literature on the Newman Penrose formalism is focused on such metrics. Our application of the formalism to the Einstein Cylinder and conformally flat manifolds is therefore a mostly unexplored research direction (as these are Petrov Type Null). This novel approach would open up new avenues for possible research in numerical relativity and the scattering problem for gravitational waves.

We examine the Einstein Cylinder as with an appropriate conformal factor, we may pull back solutions on this manifold to solutions on physical space-times [7]. Thus, understanding the scattering problem in de Sitter Space, Anti-de Sitter Space, and the class of asymptotically flat space-times. This provides a single framework that describes gravitational waves in geometries with different asymptotic structure. Numerical relativity is a highly computational endeavour. If a single simulation could be applied to a wide range of systems, it would significantly reduce the amount of computation required to study space-times with various asymptotic structures.

Furthermore, in the Newman Penrose formalism each spin coefficient describes a physical property of null congruences [8]. This provides a natural language for describing gravitational waves, as well as allowing us to focus our simulations on the properties we are most interested in. Altogether, it is clear that a successful application of the Newman Penrose formalism to the Einstein Cylinder would be a useful, and widely applicable approach to the study of gravitational radiation.

2 Background

2.1 The Einstein Cylinder

The metric for the Einstein Cylinder is

$$\begin{aligned} \mathrm{d}s^2 &= \mathrm{d}t^2 - r^2\,\mathrm{d}{\Omega_3}^2\,, \\ &= \mathrm{d}t^2 - r^2\,\mathrm{d}\chi^2 - r^2\sin^2\chi\,\mathrm{d}\theta^2 - r^2\sin^2\chi\sin^2\theta\,\mathrm{d}\phi^2\,, \\ &= g_{\mu\nu}\,\mathrm{d}x^\mu\otimes\mathrm{d}x^\nu\,. \end{aligned} \tag{2.1}$$

Here, $\mathrm{d}{\Omega_3}^2$ is the metric on a 3-sphere for which $\chi\in[0,\pi]$, $\theta\in[0,\pi]$, and $\phi\in[0,2\pi]$ are coordinates. The topology of the Einstein Cylinder, from which it takes its name, is $\mathbb{R}\times S^3$ (as an ordinary cylinder has the topology $\mathbb{R}\times S^1$) [9].

The Einstein Field Equations, given by

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = -8\pi T_{\mu\nu}, \tag{2.2}$$

relate the curvature of spacetime to the distribution of matter and energy (in units where $c = G = 1$) [10]. In these equations, $R_{\mu\nu}$ is the Ricci Tensor, R is the Ricci Scalar, Λ is the cosmological constant and $T_{\mu\nu}$ is the stress energy tensor.

The Einstein Cylinder is a model of a static universe with a uniform energy density. Asserting matter is uniformly distributed throughout the universe is done by setting

$$T_{\mu\nu} = \begin{pmatrix} \mu & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{2.3}$$

where μ is the energy density. Solving Equation (2.2) with this $T_{\mu\nu}$ and $\Lambda = 0$ results in a metric which represents a non-static universe [9]. At the time, Einstein believed an expanding (or contracting) universe to be unrealistic. This is the origin of the cosmological constant. By including the $\Lambda g_{\mu\nu}$ term in Equation (2.2) and setting

$$\Lambda = \frac{1}{r^2}, \tag{2.4}$$

one obtains the Einstein cylinder metric in Equation (2.1).

One important aspect of the Einstein Cylinder is that it is conformally flat. This means the Weyl Tensor

$$C_{abcd} = R_{abcd} - \frac{1}{2}(g_{a[c}R_{d]b} - g_{b[c}R_{d]a}) + \frac{1}{6}g_{a[c}g_{d]b}R \tag{2.5}$$

vanishes. That is,

$$C_{abc}{}^{d} = 0. \tag{2.6}$$

Under a conformal rescaling of the metric $\hat{g}_{ab} = \Omega^2 g_{ab}$ where $\Omega \neq 0$, the Weyl tensor $C_{abc}{}^{d}$ is invariant [11].

By far, the most useful modern application of the Einstein Cylinder is the fact that many other important space-times are conformally related to it [7]. Some common expamples of these conformally related metrics are Minkowski Space, de Sitter Space, and Anti-de Sitter Space. This means that understanding gravitational waves in the Einstein Cylinder will help us understand gravitational waves in many other contexts.

2.2 The Newman Penrose Formalism

The Newman Penrose Formalism [10] is an approach to general relativity that is well suited for studying gravitational waves. In this approach, one constructs vectors from two-component spinors. Consider components $V^{\mathbf{a}} = (t, x, y, z)$ of some vector in an orthonormal basis. Here, we use the convention that boldface indices represent classic, basis dependent indices, whereas the regular indices are abstract [10]. We represent this vector with the hermitian matrix

$$V^{\mathbf{a}} \equiv \begin{pmatrix} t+z & x+iy \\ x-iy & t-z \end{pmatrix}. \tag{2.7}$$

We choose this representation as the determinant of the matrix is the length of the vector $\eta_{ab}V^aV^b = t^2 - x^2 - y^2 - z^2$. Therefore, if the vector is null, the hermitian matrix is of rank 1 and admits the decomposition

$$V^{\mathbf{a}} = \begin{pmatrix} t+z & x+iy \\ x-iy & t-z \end{pmatrix} = \begin{pmatrix} \kappa^0 \\ \kappa^1 \end{pmatrix} \begin{pmatrix} \bar{\kappa}^0 & \bar{\kappa}^1 \end{pmatrix} = \kappa^A \bar{\kappa}^{A'} \tag{2.8}$$

where the bar and dashed index indicate complex conjugation [10].

With the above motivation, we choose a spin-frame $\{o^A, \iota^A\}$ and normalise it in the sense that

$$o_A \iota^A = 1. \tag{2.9}$$

Note that in two-component spinor algebra [10], indices in a normalised basis are lowered with the antisymmetric quantity

$$\varepsilon_{AB} = o_A \iota_B - \iota_A o_B, \qquad \varepsilon_{\mathbf{AB}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{2.10}$$

and raised with

$$\varepsilon^{AB} = o^A \iota^B - \iota^A o^B, \qquad \varepsilon^{\mathbf{AB}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{2.11}$$

We then define the null tetrad

$$l^a = o^A \bar{o}^{A'}, \qquad n^a = \iota^A \bar{\iota}^{A'}, \qquad m^a = o^A \bar{\iota}^{A'}, \qquad \bar{m}^a = \iota^A \bar{o}^{A'}. \tag{2.12}$$

The covariant derivative in each of the four basis vector directions $\lambda_{\mathbf{i}}^a = (l^a, n^a, m^a, \bar{m}^a)$ are given the names

$$\lambda_{\mathbf{i}}^a \nabla_a = (D, D', \delta, \delta'). \tag{2.13}$$

We then define the spin-coefficients $\gamma_{\mathbf{AA'B}}{}^{\mathbf{C}}$ analogously to the connection coefficients $\Gamma_{\mu\nu}{}^{\sigma}$. That is,

$$[\nabla_{AA'} \kappa^B]_{\text{Components}} = \nabla_{\mathbf{AA'}} \kappa^{\mathbf{B}} + \gamma_{\mathbf{AA'C}}{}^{\mathbf{B}} \kappa^{\mathbf{C}}. \tag{2.14}$$

Unique names are also given to each of the sixteen spin-coefficients as outlined in Table 1. It should be noted that if the normalisation condition in Equation (2.9) is satisfied, the spin-coefficients are such that

$$\gamma_{\mathbf{AA'BC}} = \gamma_{\mathbf{AA'CB}} \qquad \Longrightarrow \qquad \epsilon = -\gamma', \quad \gamma = -\epsilon', \quad \alpha = -\beta', \quad \beta = -\alpha'. \tag{2.15}$$

In practice, to calculate the spin coefficients, it is far easier to take linear combinations of terms of the form

$$\lambda_{\mathbf{j}}^b \lambda_{\mathbf{k}}^a \nabla_a \lambda_b^{\mathbf{i}} \tag{2.16}$$

where $\lambda_{\mathbf{i}}^a$ is the null tetrad basis [6].

Finally, the Riemann tensor is decomposed into three completely symmetric spinors Ψ_{ABCD}, $\Phi_{ABA'B'}$, and Π [10]. The spinor Ψ_{ABCD} is known as the Weyl Conformal Spinor and contains information about the Weyl curvature. Together, the spinors $\Phi_{ABA'B'}$ and Π describe the Ricci curvature. We associate these spinors with 15 scalar curvature functions as described in Table 2.

At first, individually naming all components of the connection and curvature quantities seems overly complex. However, it is very useful in some contexts. Firstly this approach is much more feasible than in a traditional vector formalism of general relativity. Using complex

$\gamma_{\mathbf{AA'B}}{}^{\mathbf{C}} =$

AA′ \ BC	00	01	10	11
00′	ϵ	$-\kappa$	$-\tau'$	γ'
10′	α	$-\rho$	$-\sigma'$	β'
01′	β	$-\sigma$	$-\rho'$	α'
11′	γ	$-\tau$	$-\kappa'$	ϵ'

Table 1: The conventional symbols used for each of the spin-coefficients. The significance of the prime notation arises from Equation (2.17). Adapted from Penrose Rindler Volume 1 [10].

Symbol	Spinor Definition	Null Tetrad Definition
Ψ_0	Ψ_{0000}	$C_{abcd}l^a m^b l^c m^d$
Ψ_1	Ψ_{0001}	$C_{abcd}l^a m^b l^c n^d$
Ψ_2	Ψ_{0011}	$C_{abcd}l^a m^b \bar{m}^c n^d$
Ψ_3	Ψ_{0111}	$C_{abcd}l^a n^b \bar{m}^c n^d$
Ψ_4	Ψ_{1111}	$C_{abcd}\bar{m}^a n^b \bar{m}^c n^d$
Φ_{00}	$\Phi_{000'0'}$	$-\frac{1}{2}R_{ab}l^a l^b$
Φ_{01}	$\Phi_{000'1'}$	$-\frac{1}{2}R_{ab}l^a m^b$
Φ_{02}	$\Phi_{001'1'}$	$-\frac{1}{2}R_{ab}m^a m^b$
Φ_{10}	$\Phi_{010'0'}$	$-\frac{1}{2}R_{ab}l^a \bar{m}^b$
Φ_{11}	$\Phi_{010'1'}$	$-\frac{1}{2}R_{ab}l^a n^b + 3\Pi$
Φ_{12}	$\Phi_{011'1'}$	$-\frac{1}{2}R_{ab}m^a n^b$
Φ_{20}	$\Phi_{110'0'}$	$-\frac{1}{2}R_{ab}\bar{m}^a \bar{m}^b$
Φ_{21}	$\Phi_{110'1'}$	$-\frac{1}{2}R_{ab}\bar{m}^a n^b$
Φ_{22}	$\Phi_{111'1'}$	$-\frac{1}{2}R_{ab}n^a n^b$
Π	Π	$\frac{1}{24}R$

Table 2: Definitions for the curvature scalars. Practically, the tetrad definition is far easier to calculate. However, the spinor definition provides the motivation behind the naming convention.

numbers has already reduced the total number of scalar quantities to consider. There is also the additional 'priming' symmetry. The priming operation is the spin-frame transformation

$$o^A \mapsto i\iota^A, \qquad \iota^A \mapsto io^A, \qquad \bar{o}^{A'} \mapsto -i\bar{\iota}^{A'}, \qquad \bar{\iota}^{A'} \mapsto -i\bar{o}^{A'}. \tag{2.17}$$

It maps all unprimed spin-coefficients and derivative operators to their primed versions [10]. Furthermore, it permutes the curvature scalars [10] as

$$\begin{aligned} &\Psi_r \mapsto \Psi_s : 0 \leftrightarrow 4, \quad 1 \leftrightarrow 3, \quad 2 \leftrightarrow 2, \\ &\Phi_{rs} \mapsto \Phi_{tu} : 0 \leftrightarrow 2, \quad 1 \leftrightarrow 1. \end{aligned}$$

Finally, if we align our spin-frame with the direction of a null congruence, each of the spin coefficients has a clear geometrical meaning. For example, ρ measures the twisting of null rays and σ measures the shearing [8].

Now recall the Ricci identity [10]

$$(\nabla_a \nabla_b - \nabla_b \nabla_a)V^d = R_{abc}{}^d V^c, \tag{2.18}$$

and Bianchi identity [10]

$$\nabla_e R_{abc}{}^d + \nabla_a R_{bec}{}^d + \nabla_b R_{eac}{}^d = 0. \tag{2.19}$$

These relate the connection coefficients $\Gamma_{\mu\nu}{}^\sigma$ to the curvature $R_{\mu\nu\sigma}{}^\lambda$. It turns out there are equivalent relations in our spinor formalism that relate the spin-coefficients to the curvature scalars. The Ricci identity in this formalism is given by the two spinor equations [10]

$$\nabla_{X'(A}\nabla_{B)}{}^{X'}\xi_C = -\Psi_{ABC}{}^D\xi_D - \Pi(\varepsilon_{AC}\xi_B + \varepsilon_{BC}\xi_A), \tag{2.20}$$

$$\nabla_{X(A'}\nabla_{B')}{}^{X}\xi_C = -\Phi_C{}^E{}_{A'B'}\xi_E. \tag{2.21}$$

The Bianchi identity is [10]

$$\nabla^A{}_{B'}\Psi_{ABCD} = \nabla_B{}^{A'}\Phi_{CDA'B'} - 2\varepsilon_{B(C}\nabla_{D)B'}\Pi. \tag{2.22}$$

These equations, when written in terms of the spin-coefficients from Table 1 and curvature scalars from Table 2 constitute the NP equations. We will refrain from writing all 26 equations in full. Instead, we will give only the required ones in Section 4.

Another insight gained from this spinor approach to general relativity is realised in Equation (2.22). In the vacuum case where $\Phi_{CDA'B'} = 0$, contracting with $\bar{\varepsilon}^{A'B'}$ yields

$$\nabla^{AA'}\Psi_{ABCD} = 0. \tag{2.23}$$

This is the field equation for a spin 2, zero rest-mass particle [10]. It is common to refer to this particle as the 'graviton'. This aspect of the conformal spinor is not directly related to the rest of the report, but serves to further motivate the use of a spinor formalism.

2.3 The Teukolsky Equation

In a traditional perturbation method [12], one considers changes to the metric of the form

$$g_{\mu\nu} \to g_{\mu\nu} + h_{\mu\nu}.$$

It's then possible to calculate to first order the perturbed connection coefficients, curvature tensors, and express them in terms of background quantities. Rather than looking at perturbations of the metric in the traditional way, Teukolsky considered perturbations of NP scalars [1]. In the case of the Kerr metric, the equation for perturbations of Ψ_0 and Ψ_4 decouple from the rest. This simplifies the problem considerably. These are the main quantities of interest as Ψ_0 and Ψ_4 represent ingoing and outgoing gravitational radiation via the peeling property [11]. Furthermore, an equation for the perturbation of Ψ_0 becomes an equation for the perturbation of Ψ_4 under the priming operation of Equation (2.17).

Perturbations are performed on each spin coefficient and curvature scalar [1] as

$$\kappa \to \kappa + \kappa^B, \qquad \sigma' \to \sigma' + \sigma'^B, \qquad \Psi_0 \to \Psi_0 + \Psi_0^B, \qquad \text{etc.} \tag{2.24}$$

where a B represents a perturbation. Then any and all terms involving two perturbation quantities are considered small and hence, ignored. Decoupling an equation for Ψ_0^B involves obtaining an equation in which there are no other non-zero perturbation terms.

The Kerr metric is well suited for analysis in the NP formalism as it is a Petrov Type D metric [1]. The Petrov classification groups space-times by the behaviour of their conformal spinors. As the conformal spinor is completely symmetric, it admits the decomposition

$$\Psi_{ABCD} = \alpha_{(A}\beta_B\gamma_C\delta_{D)}, \tag{2.25}$$

where α_A, β_B, γ_C, and δ_D are spinors representing the Principal Null Directions (PNDs) [6]. In the case of a Petrov Type D spacetime, there are two pairs of coinciding PNDs. That is, $\Psi_{ABCD} = \alpha_{(A}\alpha_B\gamma_C\gamma_{D)}$. Thus, there is a natural choice of spin-frame where we align o^A and ι^A with the PNDs of the conformal spinor. After choosing this spin-frame, one obtains that

$$\Psi_0 = \Psi_1 = \Psi_3 = \Psi_4 = 0, \tag{2.26}$$
$$\kappa = \kappa' = \sigma = \sigma' = 0. \tag{2.27}$$

This, combined with the vacuum conditions

$$\Phi_{rs} = \Pi = 0 \tag{2.28}$$

is what allows the decoupling of the equations. The final result, for the decoupled Ψ_0^B is

$$\begin{aligned}&[(D - 3\epsilon + \bar{\epsilon} - 4\rho - \bar{\rho})(D' - 4\gamma - \rho') \\ &\qquad - (\delta - \bar{\tau}' - \bar{\alpha} - 3\beta - 4\tau)(\delta' - \tau' - 4\alpha) - 3\Psi_2]\Psi_0^B = 4\pi T,\end{aligned} \tag{2.29}$$

where T is a term representing a collection of stress energy perturbations [1].

3 Connection and Curvature of the Einstein Cylinder

3.1 Connection Coefficients and Curvature Tensors

Throughout the rest of this report, Latin indices will continue to represent abstract indices, but Greek indices (μ, ν, etc.) will represent the coordinate frame. We first calculate the connection coefficients of the Einstein Cylinder using the formula for the Christoffel Symbols [10]. That is,

$$\Gamma_{\mu\nu}{}^{\sigma} = \frac{1}{2} g^{\sigma\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu}). \tag{3.1}$$

The unique non-zero connection coefficients up to symmetry in the lower two indices are

$$\begin{aligned}
\Gamma_{\theta\theta}{}^{\chi} &= -\cos\chi\sin\chi, & \Gamma_{\phi\phi}{}^{\theta} &= -\cos\theta\sin\theta, \\
\Gamma_{\phi\phi}{}^{\chi} &= -\cos\chi\sin\chi\sin^2\theta, & \Gamma_{\chi\theta}{}^{\theta} &= \cot\chi, \\
\Gamma_{\theta\phi}{}^{\phi} &= \cot\theta, & \Gamma_{\chi\phi}{}^{\phi} &= \cot\chi.
\end{aligned}$$

We now calculate the Riemann tensor using the formula for a torsion-free coordinate basis [10]. We have

$$R_{\mu\nu\sigma}{}^{\lambda} = \partial_\mu \Gamma_{\sigma\nu}{}^{\lambda} - \partial_\nu \Gamma_{\sigma\mu}{}^{\lambda} + \Gamma_{\sigma\nu}{}^{\gamma}\Gamma_{\gamma\mu}{}^{\lambda} - \Gamma_{\sigma\mu}{}^{\gamma}\Gamma_{\gamma\nu}{}^{\lambda}. \tag{3.2}$$

The unique non-zero Riemann tensor components up to symmetries are

$$\begin{aligned}
R_{\theta\chi\theta}{}^{\chi} &= \sin^2\chi, \\
R_{\phi\chi\phi}{}^{\chi} &= \sin^2\chi\sin^2\theta, \\
R_{\theta\phi\theta}{}^{\phi} &= \sin^2\chi.
\end{aligned}$$

The Ricci tensor is given by

$$R_{\mu\nu} = R_{\mu\lambda\nu}{}^{\lambda} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2\sin^2\chi & 0 \\ 0 & 0 & 0 & -2\sin^2\chi\sin^2\theta \end{pmatrix},$$

and the Ricci scalar is

$$R = g^{\mu\nu}R_{\mu\nu} = \frac{6}{r^2}.$$

Using Equation (2.5), we see that

$$C_{\mu\nu\sigma\lambda} = 0,$$

and so the space-time is conformally flat as expected. This makes the space-time Petrov Type Null [6].

3.2 Constructing a Null Tetrad

In a Petrov Type Null metric, there are no principle null directions to naturally align our spin-frame with. Instead, we will construct a simple null tetrad from an orthonormal frame. As the metric of the Einstein Cylinder is diagonal, it is easy to see that the basis

$$\hat{e}_t^a = e_t^a, \qquad \hat{e}_\chi^a = \frac{1}{a}e_\chi^a, \qquad \hat{e}_\theta^a = \frac{1}{a\sin\chi}e_\theta^a, \qquad \hat{e}_\phi^a = \frac{1}{a\sin\chi\sin\theta}e_\phi^a, \tag{3.3}$$

is orthonormal in the sense that $g_{\mu\nu}\hat{e}_i^\mu \hat{e}_j^\nu = \eta_{ij}$. The null tetrad $\lambda_{\mathbf{i}}^a$ is then chosen to be

$$l^a = \frac{1}{\sqrt{2}}(\hat{e}_t^a - \hat{e}_\chi^a), \qquad n^a = \frac{1}{\sqrt{2}}(\hat{e}_t^a + \hat{e}_\chi^a), \qquad m^a = \frac{1}{\sqrt{2}}(i\hat{e}_\theta^a - \hat{e}_\phi^a) \tag{3.4}$$

with $\bar{m}^a$ the complex conjugate of m^a. We can verify that this tetrad does fit the requirements for the NP formalism by showing that

$$g_{ab}\lambda_i^a\lambda_j^b = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \tag{3.5}$$

is the metric in the null basis [6]. In terms of the coordinate basis, we have

$$l^\mu = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -r \\ 0 \\ 0 \end{pmatrix}, \qquad n^\mu = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ r \\ 0 \\ 0 \end{pmatrix}, \qquad m^\mu = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 0 \\ ir\sin\chi \\ -r\sin\chi\sin\theta \end{pmatrix}. \tag{3.6}$$

3.3 Curvature Spinors and Spin-Coefficients

We calculate the spin-coefficients by taking linear combinations of Equation (2.16) (as described in [10]). The non-zero spin-coefficients are

$$\begin{aligned} \rho &= -\frac{\sqrt{2}}{2r\tan\chi}, & \rho' &= \frac{\sqrt{2}}{2r\tan\chi}, \\ \alpha &= -\frac{i\sqrt{2}}{4r\tan\theta\sin\chi}, & \alpha' &= \frac{i\sqrt{2}}{4r\tan\theta\sin\chi}, \\ \beta &= -\frac{i\sqrt{2}}{4r\tan\theta\sin\chi}, & \beta' &= \frac{i\sqrt{2}}{4r\tan\theta\sin\chi}. \end{aligned}$$

We already know the Weyl spinor vanishes. The remaining curvature scalars are calculated using the null tetrad definitions from Table 2. The non-zero values are

$$\begin{aligned} \Phi_{00} &= \frac{1}{2r^2}, & \Phi_{11} &= \frac{1}{4r^2}, \\ \Phi_{22} &= \frac{1}{2r^2}, & \Pi &= \frac{1}{4r^2}. \end{aligned}$$

4 Perturbation Equations

4.1 Decoupling Procedure

The relevant NP equations involving Ψ_0 are

$$\begin{aligned} &(\delta' - 4\alpha + -\tau')\Psi_0 - (D - 4\rho - 2\epsilon)\Psi_1 - 3\kappa\Psi_2 = \\ &\qquad (\delta - \bar{\tau}' - 2\bar{\alpha} - 2\beta)\Phi_{00} - (D - 2\epsilon - 2\rho')\Phi_{01} + 2\sigma\Phi_{10} - 2\kappa\Phi_{11} - \kappa'\Phi_{02}, \end{aligned} \tag{4.1}$$

$$\begin{aligned} &(D' - 4\gamma - \rho')\Psi_0 - (\delta - 4\tau - 2\beta)\Psi_1 - 3\sigma\Psi_2 = \\ &\qquad (\delta - 2\bar{\tau}' - 2\beta)\Phi_{01} - (D - 2\epsilon + 2\epsilon' - \rho')\Phi_{02} + \sigma\Phi_{00} + 2\sigma\Phi_{11} - 2\kappa\Phi_{12}, \end{aligned} \tag{4.2}$$

$$(D - \rho - \rho' - 3\epsilon + \epsilon')\sigma - (\delta - \tau - \bar{\tau}' - \alpha' - 3\beta)\kappa - \Psi_0 = 0. \tag{4.3}$$

We then perform the perturbation procedure outlined in Equation (2.24) and keep only the first order terms. The results, after substituting in the vanishing spin-coefficients and curvature scalars are

$$(\delta' - 4\alpha)\Psi_0^B - (D - 4\rho)\Psi_1^B = \\ (\delta - 2\beta - 2\beta^B - 2\bar{\alpha} - 2\bar{\alpha}^B - \bar{\tau}'^B)\Phi_{00} + (\delta - 2\beta - 2\bar{\alpha})\Phi_{00}^B - (D - 2\bar{\rho})\Phi_{01}^B - 2\kappa^B\Phi_{11}, \quad (4.4)$$

$$(D' - \rho')\Psi_0^B - (\delta - 2\beta)\Psi_1^B = \bar{\sigma}'^B\Phi_{00} + (\delta - 2\beta)\Phi_{01}^B - (D - \bar{\rho})\Phi_{02}^B + 2\sigma^B\Phi_{11}, \quad (4.5)$$

$$\Psi_0^B = (D - \rho - \bar{\rho})\sigma^B - (\delta - 3\beta - \bar{\alpha})\kappa^B. \quad (4.6)$$

If we then follow the decoupling process outlined in Teukolsky's paper [1], we obtain the equation given in Appendix A. It's omitted here for brevity, but the key aspect of this equation is that Ψ_0^B is not the only perturbed quantity. In other words, the equations are not decoupled. The coefficients of these perturbed quantities are given in Table 3.

Perturbed Quantity	**Coefficient**
κ^B	$(6\beta + 2\bar{\alpha} - 2\delta)\Phi_{11} - 2\Phi_{11}\delta$
$\bar{\tau}'^B$	$3\beta + \bar{\alpha} - \delta)\Phi_{00} - \Phi_{00}\delta$
σ^B	$(8\rho + 2\bar{\rho} - 2D)\Phi_{11} - 2\Phi_{11}D$
$\bar{\sigma}'^B$	$(4\rho + \bar{\rho} - D)\Phi_{00} - \Phi_{00}D$

Table 3: The coefficients of perturbed quantities in the equation from Appendix A. The differential operator in terms of the left is applied exclusively to the Ricci curvature term. The operators on the right are applied to the perturbed quantities of interest.

4.2 Comparison with Kerr Perturbations

If the equations do not decouple nicely for the Einstein Cylinder, it's important to consider what's preventing this. It appears that the non-vanishing Ricci curvature spinor presents problems. The Kerr metric for a rotating black hole is a vacuum solution [1]. That is,

$$\Phi_{ABA'B'} = 0. \quad (4.7)$$

It can be seen in Table 3 that the troublesome coefficients all involve these Ricci curvature terms. Furthermore, if these terms were zero, the equations would decouple. From this, we see that the vacuum nature of the Kerr space-time is a very helpful property for this kind of analysis.

There is another aspect of vacuum solutions which are important in the Teukolsky Equation. In the Teukolsky Equation, one normally asserts that the Ricci perturbation terms Φ_{rs}^B are zero [1]. In the case of the Kerr metric, this is a very reasonable assumption. If we begin with a space-time devoid of mass or energy, we don't expect small curvature perturbations to change this. In the case of the Einstein Cylinder, this is not so clear. Mathematically, there would be nothing stopping us from setting these perturbation terms to zero. However, it is more difficult to argue that gravitational waves in this space-time don't move and change the already existing mass and energy. Questions of this nature are beyond the scope of this report, but are presented to demonstrate what further obstacles lie ahead.

5 Summary

The Newman Penrose formalism can be applied to the Einstein Cylinder. However, the perturbation equations do not decouple in the same way as for the Kerr metric and other Petrov Type D solutions to the Einstein Field Equations. It's important to note that it may still be possible to decouple the equations in a useful way. Although, a creative approach which differs from the derivation of the Teukolsky Equation is required. If such a method is found, the treatment of the Ricci perturbation terms would still require careful consideration.

Another avenue for possible work could be in applying the formalism to vacuum solutions that are not Petrov Type D. The non-zero Ricci curvature of the Einstein Cylinder presented a substantial challenge, despite the spin coefficients being far simpler than the Kerr metric. Therefore, it may be possible to decouple the equations in different types of vacuum solutions. Unfortunately, desiring that the Ricci curvature is zero interferes with our goal to obtain solutions on conformally flat space-times such as de Sitter and Anti-de Sitter Space. Space-times with a vanishing Ricci tensor and vanishing Weyl tensor have a vanishing Riemann tensor. This can be proven with Equation (2.5). This would restrict us only to space-times which are locally Minkowski Space, a far cry from the many models of the universe we were hoping to explore.

Finally, an alternative approach which was not considered in this report is the use of the Compacted Spin Coefficient, or GHP formalism [10]. It may be the case that when perturbing GHP quantities, the required decoupling procedure is simpler to obtain.

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Appendix

A The Perturbed Equation

$$
\begin{aligned}
&6\,\beta^2\,\phi\mathbf{00} + 6\,\beta^2\,\phi\mathbf{00B} - 8\,\beta\,\rho\,\phi\mathbf{01B} + 6\,\beta\,\kappa\mathbf{B}\,\phi\mathbf{11} + 8\,\rho\,\sigma\mathbf{B}\,\phi\mathbf{11} - 12\,\alpha\,\beta\,\psi\mathbf{0B} + 2\,\phi\mathbf{01B}\,\mathbf{l}[\beta] - 2\,\phi\mathbf{11}\,\mathbf{l}[\sigma\mathbf{B}] + \\
&5\,\beta\,\mathbf{l}[\phi\mathbf{01B}] - 4\,\rho\,\mathbf{l}[\phi\mathbf{02B}] - 2\,\sigma\mathbf{B}\,\mathbf{l}[\phi\mathbf{11}] + \mathbf{l}[\mathbf{l}[\phi\mathbf{02B}]] + \mathbf{l}[\mathbf{n}[\psi\mathbf{0B}]] - \phi\mathbf{02B}\,\mathbf{l}[\overline{\rho}] - \phi\mathbf{00}\,\mathbf{l}[\overline{\sigma\mathbf{B}^*}] - \\
&\psi\mathbf{0B}\,\mathbf{l}[\rho^*] - \mathbf{l}[\delta[\phi\mathbf{01B}]] - 4\,\rho\,\mathbf{n}[\psi\mathbf{0B}] + 8\,\beta\,\phi\mathbf{00}\,\overline{\alpha} + 8\,\beta\,\phi\mathbf{00B}\,\overline{\alpha} + 2\,\kappa\mathbf{B}\,\phi\mathbf{11}\,\overline{\alpha} - 4\,\alpha\,\psi\mathbf{0B}\,\overline{\alpha} + \\
&\mathbf{l}[\phi\mathbf{01B}]\,\overline{\alpha} + 2\,\phi\mathbf{00}\,\overline{\alpha}^2 + 2\,\phi\mathbf{00B}\,\overline{\alpha}^2 - 8\,\beta\,\phi\mathbf{01B}\,\overline{\rho} + 4\,\rho\,\phi\mathbf{02B}\,\overline{\rho} + 2\,\sigma\mathbf{B}\,\phi\mathbf{11}\,\overline{\rho} - 2\,\mathbf{l}[\phi\mathbf{02B}]\,\overline{\rho} - \mathbf{n}[\psi\mathbf{0B}]\,\overline{\rho} - \\
&2\,\phi\mathbf{01B}\,\overline{\alpha}\,\overline{\rho} + \phi\mathbf{02B}\,\overline{\rho}^2 + 4\,\rho\,\phi\mathbf{00}\,\overline{\sigma\mathbf{B}^*} - \mathbf{l}[\phi\mathbf{00}]\,\overline{\sigma\mathbf{B}^*} + \phi\mathbf{00}\,\overline{\rho}\,\overline{\sigma\mathbf{B}^*} + 3\,\beta\,\phi\mathbf{00}\,\overline{\tau\mathbf{B}^*} + \phi\mathbf{00}\,\overline{\alpha}\,\overline{\tau\mathbf{B}^*} + \\
&4\,\rho\,\psi\mathbf{0B}\,\rho^* - \mathbf{l}[\psi\mathbf{0B}]\,\rho^* + \psi\mathbf{0B}\,\overline{\rho}\,\rho^* + 4\,\psi\mathbf{0B}\,\delta[\alpha] - 2\,\phi\mathbf{00}\,\delta[\beta] - 2\,\phi\mathbf{00B}\,\delta[\beta] - 2\,\phi\mathbf{11}\,\delta[\kappa\mathbf{B}] - \\
&5\,\beta\,\delta[\phi\mathbf{00}] - 3\,\overline{\alpha}\,\delta[\phi\mathbf{00}] - \overline{\tau\mathbf{B}^*}\,\delta[\phi\mathbf{00}] - 5\,\beta\,\delta[\phi\mathbf{00B}] - 3\,\overline{\alpha}\,\delta[\phi\mathbf{00B}] + 4\,\rho\,\delta[\phi\mathbf{01B}] + \\
&3\,\overline{\rho}\,\delta[\phi\mathbf{01B}] - 2\,\kappa\mathbf{B}\,\delta[\phi\mathbf{11}] + 4\,\alpha\,\delta[\psi\mathbf{0B}] - \delta[\mathbf{l}[\phi\mathbf{01B}]] - 2\,\phi\mathbf{00}\,\delta[\overline{\alpha}] - 2\,\phi\mathbf{00B}\,\delta[\overline{\alpha}] + \\
&2\,\phi\mathbf{01B}\,\delta[\overline{\rho}] - \phi\mathbf{00}\,\delta[\overline{\tau\mathbf{B}^*}] + \delta[\delta[\phi\mathbf{00}]] + \delta[\delta[\phi\mathbf{00B}]] - \delta[\delta^*[\psi\mathbf{0B}]] + 3\,\beta\,\delta^*[\psi\mathbf{0B}] + \overline{\alpha}\,\delta^*[\psi\mathbf{0B}]
\end{aligned}
$$

Figure 1: The decoupled expression after applying the decoupling procedure to the Einstein Cylinder. One obtains the equation by setting this expression equal to zero. The calculation was performed in Mathematica by adapting a notebook provided by Chris Stevens. The operators l and n correspond to D and D' respectively. The asterisk is the priming operation from Equation (2.17). All terms named with a B are perturbed quantities.

References

[1] Saul A Teukolsky. "Perturbations of a rotating black hole. I. Fundamental equations for gravitational, electromagnetic, and neutrino-field perturbations". In: *Astrophysical Journal, Vol. 185, pp. 635-648 (1973)* 185 (1973), pp. 635–648.

[2] Albert Einstein. "The field equations of gravitation". In: *Sitzungsber. Preuss. Akad. Wiss. Berlin (Math. Phys.)* 1915 (1915), pp. 844–847.

[3] Albert Einstein. "Approximative integration of the field equations of gravitation". In: *Sitzungsber. Preuss. Akad. Wiss. Berlin (Math. Phys.)* 1916.688-696 (1916), p. 1.

[4] B. P. Abbott et al. "Observation of Gravitational Waves from a Binary Black Hole Merger". In: *Physical Review Letters* 116.6 (Feb. 2016). ISSN: 1079-7114. DOI: `10.1103/physrevlett.116.061102`. URL: `http://dx.doi.org/10.1103/PhysRevLett.116.061102`.

[5] Tullio Regge and John A Wheeler. "Stability of a Schwarzschild singularity". In: *Physical Review* 108.4 (1957), p. 1063.

[6] Ezra Ted Newman and Roger Penrose. "Spin-coefficient formalism". In: *Scholarpedia* 4.6 (2009), p. 7445.

[7] Stephen W Hawking and George FR Ellis. *The large scale structure of space-time*. Cambridge university press, 2023.

[8] Roger Penrose and Wolfgang Rindler. *Spinors and space-time: Volume 2, Spinor and twistor methods in space-time geometry*. Vol. 2. Cambridge University Press, 1984.

[9] Albert Einstein. "Cosmological Considerations on the General Theory of Relativity". In: (1917).

[10] Roger Penrose and Wolfgang Rindler. *Spinors and space-time*. Vol. 1. Cambridge university press, 1984.

[11] Jörg Frauendiener. "Conformal infinity". In: *Living Reviews in Relativity* 7.1 (2004). Accessed 27/01/2026. URL: `https://link.springer.com/article/10.12942/lrr-2000-4`.

[12] Kip S Thorne, Charles W Misner, and John Archibald Wheeler. *Gravitation*. Freeman San Francisco, 2000.